

Zikang Leng

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🏠 [zikangleng.github.io](https://github.com/zikangleng)

in [zikang-leng](https://github.com/zikang-leng)

🔗 [ZikangLeng](#)

🎓 [Scholar](#)

EDUCATION

Georgia Institute of Technology

Atlanta, GA

PhD in Computer Science

Aug. 2024 - Expected May. 2028

- Advised by Prof. Thomas Plötz. Research areas: human activity recognition, multimodal sensing, data generation, LLM agents

Georgia Institute of Technology

Atlanta, GA

B.S. in Computer Science (Theory & AI), B.S. in Physics | GPA: 4.0

Aug. 2021 - May. 2024

EXPERIENCE

Graduate Research Assistant

Aug. 2024 - Present

Georgia Institute of Technology

Atlanta, GA

- Developed **AgentSense** (AAAI '26): LLM-guided agents simulated **18** personas across **22** home layouts to generate **250 days** of privacy-preserving smart-home sensor data; pretraining on it beat baselines on **5** real HAR datasets.
- Built **EduGage** (IMWUT '26), fusing **8** synchronized physiological, motion, and eye-tracking signals from 16 participants to reach **83.18%** within-1 engagement accuracy, outperforming foundation-model and LLM baselines.
- Benchmarked virtual IMU generation against classical augmentation over **100** activities and **22** body locations, across 3 HAR benchmarks and 4 model architectures.

Machine Learning Engineer Intern

Aug. 2026 – Dec. 2026

BehaviorAI

Brooklyn, NY

- Developing behavior foundation models and generative methods for human behavior understanding.

Research Intern, Sensor Machine Learning

Jun. 2025 – Aug. 2025

Nokia Bell Labs

Cambridge, United Kingdom

- Fused IMU and PPG by signal quality at runtime for cardiovascular sensing, distilling from ECG at training.
- Self-supervised SwAV clustering for cross-modal distillation; assignment entropy gives uncertainty (**UK patent**).

Undergraduate Research Assistant

Aug. 2022 – May 2024

Georgia Institute of Technology

Atlanta, GA

- Developed **IMUGPT** (IMWUT '24), generating virtual IMU data from text via LLMs and motion synthesis; motion filtering and diversity metrics cut generation effort by **over 50%**.
- Evaluated alternative data-collection strategies for wheelchair transportation mode detection, reducing reliance on recruiting wheelchair users (Best Paper Nomination, ISWC '24).
- Built a personalized emotion-recognition system using wearable IMUs and virtual motion data for pretraining.
- Built **FingerSpeller** (ASSETS '23), a camera-free ASL fingerspelling system reaching **91%** character accuracy from smart rings; released a 42,288-letter dataset.

Machine Learning Research Intern – SULI

May 2023 – Aug. 2023

Argonne National Laboratory

Lemont, IL

- Accelerated GNN training for ocean simulation by **213×** using distributed training across **256 GPUs**.
- Expanded GNN training data via inverse-distance-weighted interpolation for larger-scale ocean simulation.

SELECTED PUBLICATIONS

EduGage: A Multimodal Dataset and Benchmark for Sensor-Based Momentary Assessment of Engagement in Self-Guided Video Learning

Zikang Leng*, Edan Eyal*, Yingtian Shi, Jiaman He, Yaqi Liu, Thomas Plötz

UbiComp/IMWUT '26
[arxiv][code][dataset][project]

AgentSense: Virtual Sensor Data Generation Using LLM Agents in Simulated Home Environments

Zikang Leng*, Megha Thukral*, Yaqi Liu*, Hrudhai Rajasekhar, Shruthi K Hiremath, Jiaman He, Thomas Plötz

AAAI '26
[paper][code][project]

StitchSense: Smart-Garment Sensing as a Source Domain for Watch-Based Activity Recognition

Ishita Datta, Zikang Leng, Rachit Bhayana, Thomas Plötz

UbiComp/ISWC '26
[paper]

Characterising Topic Familiarity and Query Specificity Using Eye-Tracking Data

Jiaman He, Zikang Leng, Dana McKay, Johanne R Trippas, Damiano Spina

SIGIR '25
[paper][code]

*Equal Contribution

Scaling Human Activity Recognition: A Comparative Evaluation of Synthetic Data Generation and Augmentation Techniques

Zikang Leng, Archith Iyer, Thomas Plötz

Under Review

[\[arxiv\]](#)

IMUGPT 2.0: Language-Based Cross Modality Transfer for Sensor-Based Human Activity Recognition

Zikang Leng, Amitrajit Bhattacharjee, Hrudhai Rajasekhar, Lizhe Zhang, Elizabeth Bruda, Hyeokhyen Kwon, Thomas Plötz

UbiComp/IMWUT '24

[\[paper\]](#)[\[code\]](#)[\[project\]](#)

More Data for People with Disabilities! Comparing Data Collection Efforts for Wheelchair Transportation Mode Detection (**Best Paper Nomination**)

Sungjin Hwang, Zikang Leng*, Seungwoo Oh, Kwanguk Kim, Thomas Plötz*

UbiComp/ISWC '24

[\[paper\]](#)

Emotion Recognition on the Go: Utilizing Wearable IMUs for Personalized Emotion Recognition

Zikang Leng, Myeongul Jung*, Sungjin Hwang, Seungwoo Oh, Lizhe Zhang, Thomas Plötz, Kwanguk Kim*

UbiComp'24 HASCA

[\[paper\]](#)

FingerSpeller: Camera-Free Text Entry Using Smart Rings for American Sign Language Fingerspelling Recognition

David Martin, Zikang Leng*, Tan Gemicioglu, Jon Womack, Jocelyn Heath, Bill Neubauer, Hyeokhyen Kwon, Thomas Plötz, Thad Starner*

ASSETS '23

[\[paper\]](#)

Generating Virtual On-body Accelerometer Data from Virtual Textual Descriptions for Human Activity Recognition (**Best Paper Honorable Mention**)

Zikang Leng, Hyeokhyen Kwon, Thomas Plötz

UbiComp/ISWC '23

[\[paper\]](#)[\[code\]](#)[\[project\]](#)

[\[news\]](#)

On the Utility of Virtual On-body Acceleration Data for Fine-grained Human Activity Recognition

Zikang Leng, Yash Jain, Hyeokhyen Kwon, Thomas Plötz

UbiComp/ISWC '23

[\[paper\]](#)

PATENTS

Language-based Cross-modality Transfer System for Generating Virtual Data to Enhance Activity Recognition

Zikang Leng, Amitrajit Bhattacharjee, Hrudhai Rajasekhar, Lizhe Zhang, Elizabeth Bruda, Hyeokhyen Kwon, Thomas Plötz

US 2026/0289407 A1

A System for Quality-Aware Multimodal Fusion Representation Learning with Uncertainty Estimation for Health Condition Modelling

Zikang Leng, Mohammad Malekzadeh, Chi Ian Tang

UK Patent Filed '25

AWARDS

NSF Graduate Research Fellowship (GRFP) 2024

- Three-year \$37,000 annual stipend plus \$16,000 yearly tuition allowance for outstanding STEM graduate students

Georgia Tech President's Fellowship 2024

- Awarded to select doctoral students for exemplary scholarship and innovation. \$5,500 annual stipend, renewable for three years

Georgia Tech Institute Fellowship 2024

- Georgia Tech's most prestigious graduate fellowship, awarded to a select few President's Fellowship holders. \$2,000 annual stipend

Provost's Academic Excellence Award 2024

- Top graduating senior from Georgia Tech's College of Computing. \$2,000 award

President's Undergraduate Research Awards (Salary and Travel) 2023

- \$2,500 to conduct undergraduate research with Georgia Tech faculty, plus a travel grant

TECHNICAL SKILLS

Languages: Python, C/C++, Java, Bash, LaTeX

ML & Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, NumPy, OpenCV, Ray Tune

Infrastructure: Distributed multi-GPU training, CUDA, Docker, Git

MENTORED STUDENTS

Mentored 5 masters and 5 undergraduate students; 7 are co-authors on the publications above.